

Task 6: Obstacle detection and Robot Control

Prerequisites

- Verify the Thonny(Python IDE) program in the Raspberry pi.
- Basic Knowledge on Ultrasonic Sensor, motor Control and graph plotting using Python required.

Preparation

- View the example Python scripts to see how to control the robots movement using the data from the Ultrasonic sensor on the Robot.
- You can find a reference here:
<https://circuitdigest.com/microcontroller-projects/raspberry-pi-obstacle-avoiding-robot>

Task Description:

- a) Using Python use the Ultrasonic sensor to avoid collision between the robot and its surroundings while the robot moves around in random directions.
- b) Using python, Create a coordinate trace of your robots path.